

In this Standard the comma has been adopted as the decimal sign.

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NOT APPLICABLE TO NEW DRAWINGS. FOR NEW ONES SEE:  
-NE03-071 FOR GENERAL TOLERANCES OF MACHINED METALIC DETAILS;  
-NE03-080 FOR GENERAL TOLERANCES OF METAL SHEET FORMING PARTS; AND  
-FOR ASSEMBLIES INDICATE SPECIFIC TOLERANCES ON DRAWINGS.

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DETAIL PARTS AND ASSEMBLIES  
GENERAL TOLERANCES

REV DATA  
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NE 03-063

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PAGE 1 of 20



## 1. SCOPE

This Standard establishes general tolerances (dimensional and geometric) applicable to the manufacture of detail parts and assemblies of Embraer aircraft.

## 2. APPLICATION

This Standard is applicable whenever called out on production drawings (refer to 6.1) defining tolerances for the dimensions without specific tolerance indications.

### NOTE

This Standard shall not be called out on new machined part drawings - for these new drawings use the NE 03-071 (NI 641).

## 3. DEFINITIONS

### 3.1 GENERAL TOLERANCE

Tolerance that is common to several dimensions or geometric configurations. It is not indicated next to nominal dimension on the drawing.

### 3.2 SPECIFIC TOLERANCE

Tolerance indicated next to the nominal dimension on the drawing.

### 3.3 ASSEMBLY TOLERANCE

General or specific tolerance which limits the positioning of components in a rigid assembly (structural assemblies, etc).

### NOTE

For the purpose of this Standard, components adjustment or riging tolerances are not considered as assembly tolerances.

## 4. APPLICABLE DOCUMENTS

For the application of this Standard, it is necessary to refer to the documents listed below and in Table 6 (item 5.3).

### Embraer Standards

NE 03-004 - Surface Roughness.

|           |                        |           |                  |
|-----------|------------------------|-----------|------------------|
| REV.<br>E | DATA<br>DATE 24 Jan 94 | NE 03-063 | PÁGINA<br>PAGE 2 |
|-----------|------------------------|-----------|------------------|

5. REQUIREMENTS

5.1 DETAIL PART TOLERANCES

CAUTION

- The tolerances listed in the following sub-items are applicable only when the case is not mandatory (5.3, Table 7).
- The reference dimensions shall not be checked when the part is inspected. They are represented by the dimension value, followed by "REF". They have no tolerance.

5.1.1 MACHINING - GENERAL DIMENSIONS

Tolerances shall be in accordance with Table 1.

NOTES

- 1) For radii and chamfer heights refer to 5.1.2.
- 2) For holes made with drill refer to 5.1.3.

TABLE 1 : GENERAL TOLERANCE - MACHINING  
DIMENSIONS IN GENERAL

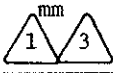
| NOMINAL LINEAR<br>DIMENSION "D"<br>mm | TOLERANCE<br>mm<br> | NOMINAL ANGULAR<br>DIMENSION "θ"<br>INCLUDING 90°,<br>NOT LISTED | TOLERANCE<br> |
|---------------------------------------|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| D < 1,5                               | +0,3<br>-0,1                                                                                           | 0° < θ < 360°                                                    | ± 0° : 30'                                                                                         |
| 1,5 ≤ D ≤ 100                         | ± 0,3                                                                                                  |                                                                  |                                                                                                    |
| 100 < D ≤ 400                         | ± 0,4                                                                                                  |                                                                  |                                                                                                    |
| 400 < D ≤ 1000                        | ± 0,6                                                                                                  |                                                                  |                                                                                                    |
| 1000 < D ≤ 2000                       | ± 0,8                                                                                                  |                                                                  |                                                                                                    |
| 2000 < D < -                          | ± 0,000 6 D                                                                                            |                                                                  |                                                                                                    |

 Refer to 5.3.

### 5.1.2 MACHINING - RADII AND CHAMFER HEIGHTS

The tolerances for radii and chamfers (inner and outer) shall be in accordance with Table 2.

TABLE 2 : GENERAL TOLERANCE - MACHINING OF RADII AND CHAMFERS

| NOMINAL SIZE<br>RANGE IN mm<br>"R" = RADIUS<br>"H" = CHAMFER                                         | TOLERANCE<br> | "MPD" OF FILLET <br>BETWEEN INNER RADII<br>AND ADJACENT SURFACES |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
|  0,5 < R (ou H) ≤ 3 | ± 0,3                                                                                          | SHALL BE IN<br>ACCORDANCE<br>WITH TABLE 4                                                                                                           |
| 3 < R (ou H) ≤ 6                                                                                     | ± 0,5                                                                                          |                                                                                                                                                     |
| 6 < R (ou H) -                                                                                       | ± 1,0                                                                                          |                                                                                                                                                     |

 Refer to 5.3.

 a) sharp corners not specified on production drawings, shall have  $R \leq 0,5 \text{ mm}$  or  $H \leq 0,5 \text{ mm} \times 45^\circ$ .  
b)  $R$  (or  $H$ )  $\leq 0,5 \text{ mm}$ , shown without tolerance on production drawings, shall be considered as  $R$  (or  $H$ ) max.

 Chamfer angle tolerance:  $\pm 1^\circ$ .

 MPD - Maximum Permissible Deviation.

### 5.1.3 MACHINING - DRILLED HOLES

Tolerances for diameter and angular displacement of the hole shall be in accordance with Table 3.

TABLE 3 : GENERAL TOLERANCE - DRILLED HOLES

| NOMINAL HOLE<br>DIAMETER "φ"<br>mm  | TOLERANCE<br>"φ"<br>mm  | HOLE ANGULAR DISPLACEMENT<br>TOLERANCE  |               |
|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------|
|                                                                                                                        |                                                                                                            | STATIC<br>DRILL                                                                                                              | HAND<br>DRILL |
| - φ ≤ 3                                                                                                                | +0,14<br>0                                                                                                 | ± 1°                                                                                                                         | ± 3°          |
| 3 < φ ≤ 6                                                                                                              | +0,18<br>0                                                                                                 |                                                                                                                              |               |
| 6 < φ ≤ 10                                                                                                             | +0,22<br>0                                                                                                 |                                                                                                                              |               |
| 10 < φ ≤ 18                                                                                                            | +0,27<br>0                                                                                                 |                                                                                                                              |               |
| 18 < φ ≤ 30                                                                                                            | +0,33<br>0                                                                                                 |                                                                                                                              |               |
| 30 < φ -                                                                                                               | REFER TO H13 IN<br>NBR 6158 (NB 86)                                                                        |                                                                                                                              |               |

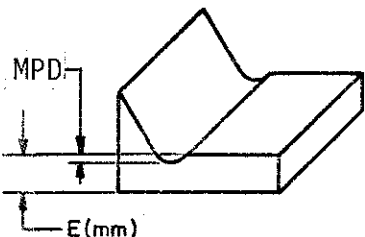
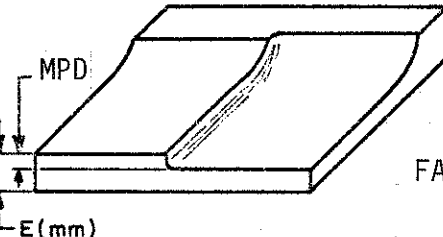
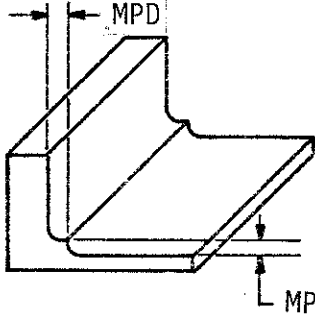
 Refer to 5.3

- 2

#### 5.1.4 MACHINING - MAXIMUM PERMISSIBLE DEVIATIONS

Maximum permissible deviations shall be in accordance with Table 4.

TABLE 4 : GENERAL TOLERANCE - MAXIMUM PERMISSIBLE DEVIATIONS  
IN MACHINED SURFACES

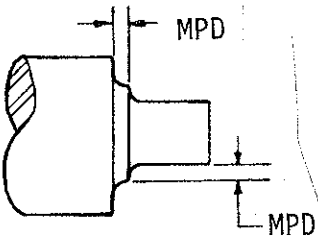
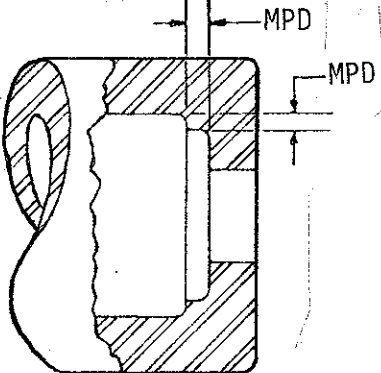
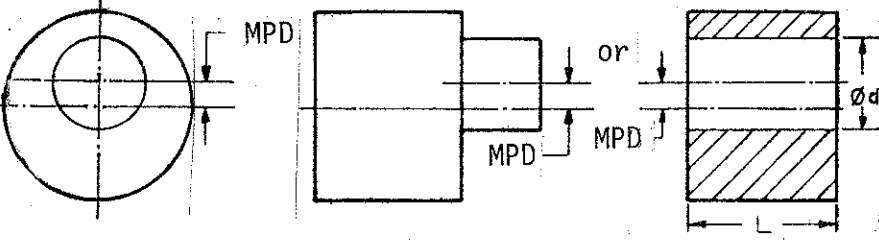
| DESCRIPTION                                                                                                                                        | MPD IN mm,<br>UNLESS OTHERWISE<br>INDICATED<br><div style="display: flex; justify-content: space-around; align-items: center;"> <span style="border: 1px solid black; padding: 2px 5px;">1</span> <span style="border: 1px solid black; padding: 2px 5px;">2</span> </div> |
|----------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>FACING OF TWO INTERSECTING SURFACES (MAXIMUM DEPRESSION)</p> | <div style="border-bottom: 1px solid black; padding-bottom: 5px;">E/10 FOR <math>E \leq 3</math></div> <div style="padding-top: 5px;">0,3 FOR <math>E &gt; 3</math></div>                                                                                                  |
|  <p>FACING OF A SINGLE SURFACE (MAXIMUM MISMATCH)</p>           | <div style="border-bottom: 1px solid black; padding-bottom: 5px;">E/10 FOR <math>E \leq 3</math></div> <div style="padding-top: 5px;">0,3 FOR <math>E &gt; 3</math></div>                                                                                                  |
|  <p>FACING OF TWO INTERSECTING SURFACES (RADIUS MISMATCH)</p>   | <div style="padding-top: 5px;">0,5</div>                                                                                                                                                                                                                                   |

continued

- 

TABLE 4 : GENERAL TOLERANCE - MAXIMUM PERMISSIBLE DEVIATIONS  
IN MACHINED SURFACES

continued

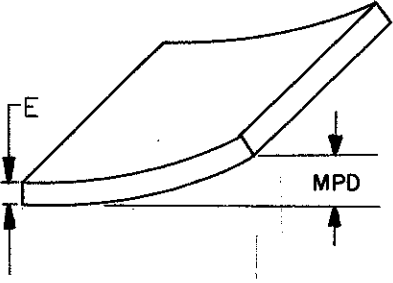
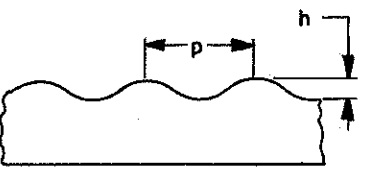
| DESCRIPTION                                                                                                                                                                                           | MPD IN mm,<br>UNLESS OTHERWISE<br>INDICATED <span style="border: 1px solid black; padding: 2px;">1</span> <span style="border: 1px solid black; padding: 2px;">2</span>                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>EXTERNAL MACHINING<br/>(MAXIMUM OVERHANG<br/>OR DEPRESSION)</p>                                                  | <p>+ 0,2<br/>- 0,1</p>                                                                                                                                                                                                                   |
|  <p>INTERNAL MACHINING<br/>(MAXIMUM OVERHANG)</p>                                                                   | <p>0,5</p>                                                                                                                                                                                                                               |
| <p>MAXIMUM COAXIALITY DEVIATION</p> <p>EXTERNAL COAXIAL MACHINING      EXTERNAL X INTERNAL COAXIAL MACHINING</p>  | <p>EXTERNAL MACHINING:<br/>0,05</p> <p>INTERNAL OR INTERNAL X<br/>EXTERNAL MACHINING :</p> <p>0,05 FOR <math>\frac{L}{\phi d} \leq 2</math><br/> 0,025 x <math>\frac{L}{\phi d}</math> FOR<br/> <math>\frac{L}{\phi d} &gt; 2</math></p> |

- 1 MPD - Maximum Permissible Deviation.  
2 Refer to 5.3.

continued

TABLE 4 : GENERAL TOLERANCE - MAXIMUM PERMISSIBLE DEVIATIONS  
IN MACHINED SURFACES

continued

| DESCRIPTION                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | MPD IN mm<br>(UNLESS INDICATED OTHERWISE) 1/2                                                                       |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
|  <p>MAXIMUM DISTORTION (WARP) OF EDGES AND/OR SURFACES AFTER MACHINING DUE TO HEAT TREATMENT RESIDUAL TENSION AND/OR IMPROPER ATTACHMENT OF WORKPIECE TO THE MACHINE.<br/>EDGES : STRAIGHTNESS ( — ) OR ANY LINE SHAPE ( ⤿ )<br/>SURFACES : FLATNESS ( ▭ ) OR ANY SURFACE SHAPE ( ⤿ ).</p> <p>THE MAXIMUM DISTORTION RESULTED FROM WARP SHALL BE MEASURED AFTER APPLYING LOCALIZED WEIGHTS TO FIT THE PART INTO ITS CHECKING DEVICE, ACCORDING TO :</p> <p>1) 1 kgf FOR E ≤ 1 mm<br/>2) 2 kgf FOR E &gt; 1 mm</p> <p>THE WEIGHTS SHALL BE APPLIED AT 300 mm INTERVALS OR AT EQUIVALENT SMALLER INTERVALS (EXAMPLES : 1/2 OF THE SPECIFIED WEIGHT AT 150 mm INTERVALS OR 1/4 OF THE SPECIFIED WEIGHT AT 75 mm INTERVALS)</p> | <p>0,8 MAX</p>                                                                                                      |
|  <p>MAXIMUM SURFACE ROUGHNESS</p> <p>NOTE : MEASURE Ra IN ACCORDANCE WITH NE 03-004.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Ra = 3,2 μm</p>                                                                                                  |
|  <p>MAXIMUM SURFACE WAVINESS</p> <p><math>h \approx 20 Ra</math><br/><math>p \approx 10 h</math>.</p> <p>NOTE<br/>WAVINESS SHALL BE CONSIDERED WITHIN A 13 mm RANGE, MEASURED IN A DIRECTION PARALLEL WITH p.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <p>STEELS :</p> <p><math>h = 0,1 ; p = 1,0</math>.</p> <p>OTHER ALLOYS :</p> <p><math>h = 0,06 ; p = 0,6</math></p> |

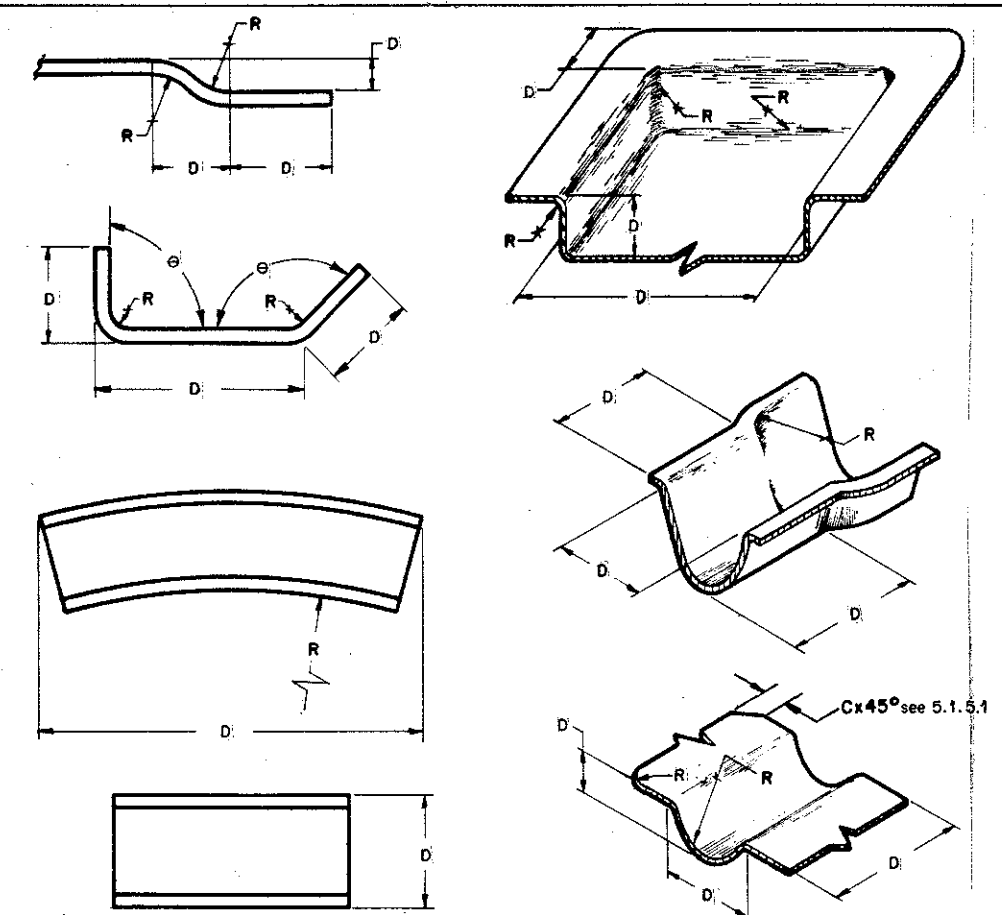
1 MPD - Maximum Permissible Deviation.

2 Refer to 5.3.

5.1.5 SHEET FORMING

General tolerance shall be according to Table 5 for linear and angular, dimensions, unless otherwise specified.

TABLE 5 : GENERAL TOLERANCES SHEET FORMING



| NOMINAL LINEAR<br>DIMENSION "D"<br>OR "R" | TOLERANCE<br>mm  | NOMINAL ANGULAR<br>DIMENSION "θ" (INCLUD<br>ING 90°, NOT LISTED) | TOLERANCE<br> |
|-------------------------------------------|-----------------------------------------------------------------------------------------------------|------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| - D ≤ 300                                 | ± 0,8                                                                                               | 0° < φ < 360°                                                    | ± 5 °                                                                                              |
| 300 < D ≤ 1000                            | ± 1,0                                                                                               |                                                                  |                                                                                                    |
| 1000 < D -                                | ± 0,001 2D                                                                                          |                                                                  |                                                                                                    |

-  Exceptions:
- a) see 5.3;
  - b) the general tolernace for bending radii  $R \leq 50$  mm in cold bending of metal sheets, shall be on the plus side only. Example:  $R = 30$  mm shall vary from 30,0 mm to 30,8 mm;
  - c) the general tolerance for parts with all surfaces class IV (NE 03-040) shall be  $\pm 2$  mm;
  - d) the general tolerance of flange height of parts bent on hydroform press or rubber press shall be:  
+ 1,4 mm.  
- 1,0 mm.

### 5.1.5.1 SHOP CORNER OPTIONS OF SHEET PARTS

If drawing does not indicate the note "NLXXX - CONER MUST BE AS SHOWB", shop may change corners according to options specified in Figure 1.

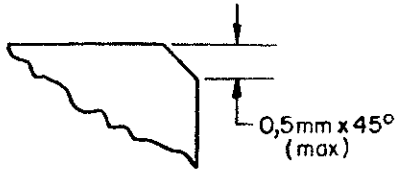
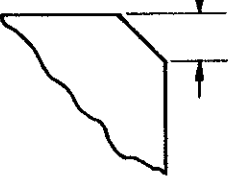
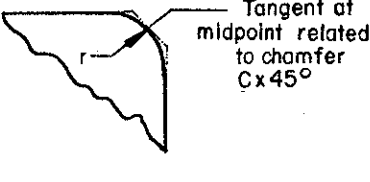
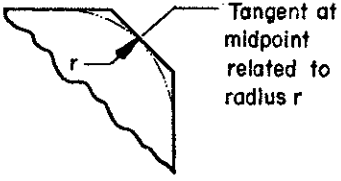
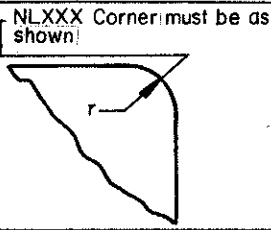
| SHOWN ON<br>DRAWING                                                                                                       | SHOP OPTION (S)                                                                                                                                         |                                                                                                                             |
|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
|                                          |  $r=0,5\text{mm}$<br>(max)                                             |  $0,5\text{mm} \times 45^\circ$<br>(max) |
|  $C \times 45^\circ$                    |  Tangent at<br>midpoint related<br>to chamfer<br>$C \times 45^\circ$ | _____                                                                                                                       |
|  $r$                                   |  Tangent at<br>midpoint<br>related to<br>radius $r$                  | _____                                                                                                                       |
|  NLXXX Corner must be as<br>shown! $r$ | _____                                                                                                                                                   | _____                                                                                                                       |

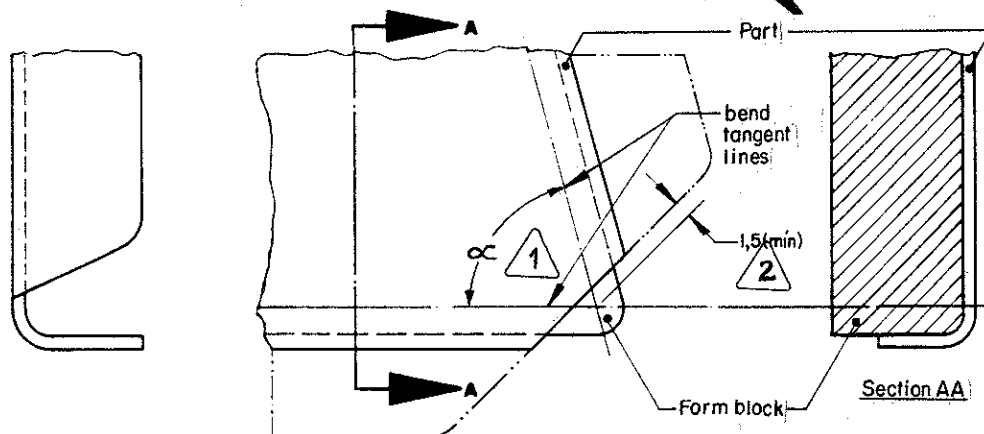
FIGURE 1 : SHOP CORNER OPTIONS

### 5.1.5.2 RADII RELIEF

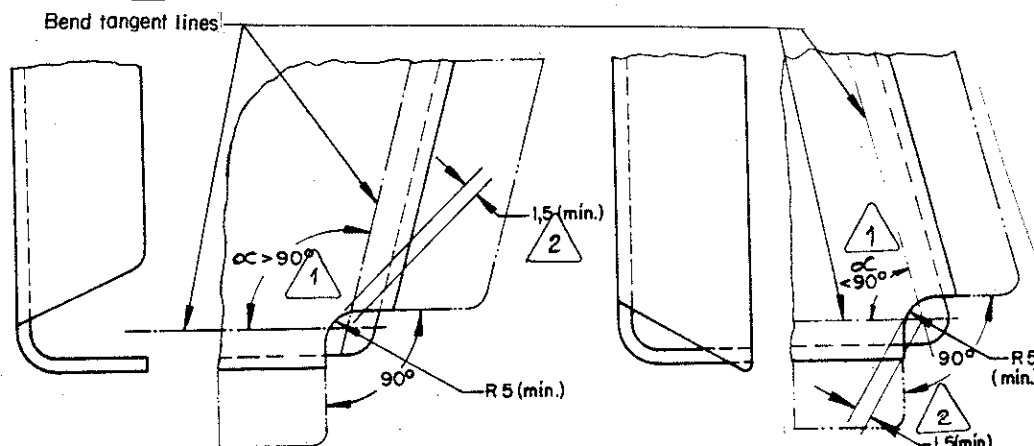
Although other way specified on drawing, sheet parts containing corners formed by two convergent flanges presenting problems upon forming such as bending radius smaller than the minimum speciied value, cracks, superficial deformations, etc shall present a radius relief according to one of the configurations specified in Figure 2.

#### NOTES

- 1) Radius relief shall not interfere in the functional performance of part (reduction of edge distance for rivet installation, reduction of mechanical strength, etc). In case of doubt, consult Structure Engineering.
- 2) Class IV parts according to NE 03-040 do not require radius relief.



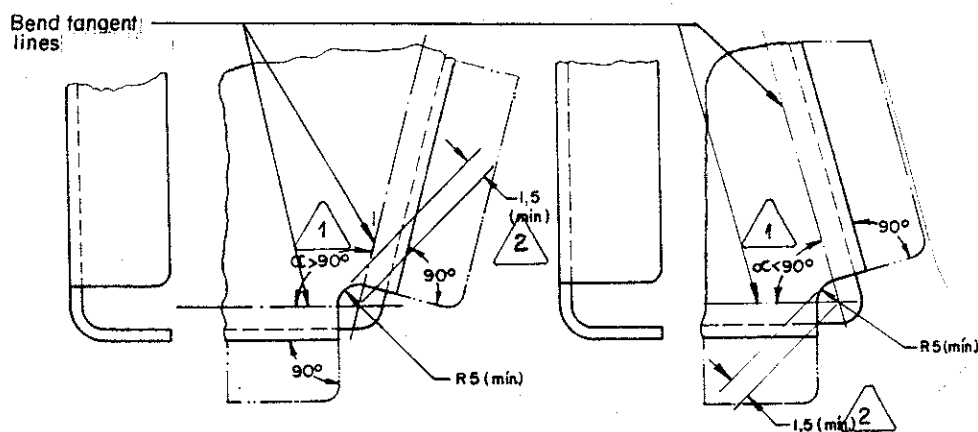
1st CONFIGURATION: STRAIGHT CUT OUT ON BLANK



NOTE

It shall be considered, in this case, the final position of bent flange if the angle  $\alpha$  is different from  $90^\circ$ .

2nd CONFIGURATION : CUT OUT UNDER ANGLE  $90^\circ$  ON BLANK



NOTE

This cut out promotes flange corners with  $90^\circ$ , independent of angle  $\alpha$ .

3rd CONFIGURATION: PERPENDICULAR CUT OUT TO THE BEND TANGENT LINES ON BLANK

- 1 Angle  $\alpha$  may be equal or different from  $90^\circ$  for all three configurations.
- 2 The maximum value ( $\approx 2,5$  mm) is free, but it shall not interfere on functional performance of part (edge distance, etc).

FIGURE 2 : RADIUS RELIEF

### 5.1.5.3 LUGS CUTOUT

#### a) General rule

The symbol  in lug representation on drawings means that it shall be cutout according to part shape. See Figure 3.

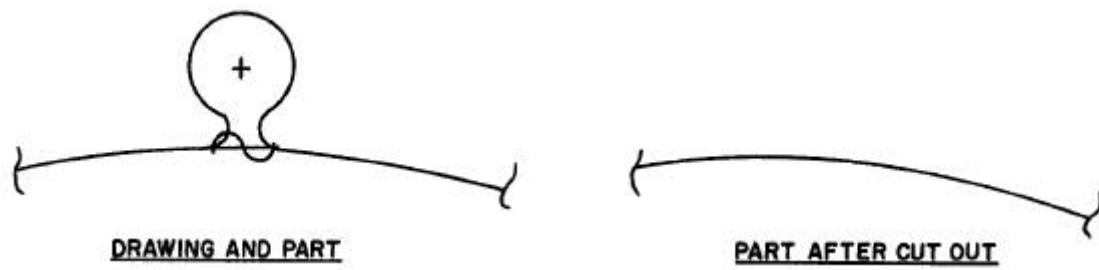


FIGURE 3: LUG CUTOUT GENERAL RULE

#### b) Lug on Winding Shape

In this case the lug cutout shall be always straight, according to indicated on Figure 4.

The part is authorized to have different contour from that indicated on drawing, in the lug (cutout) region.

#### b1) Lug indicated on Drawing

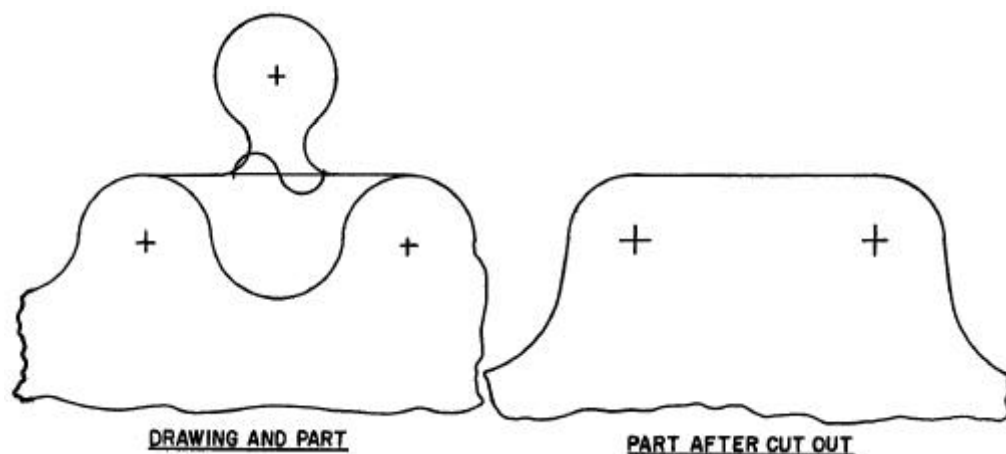
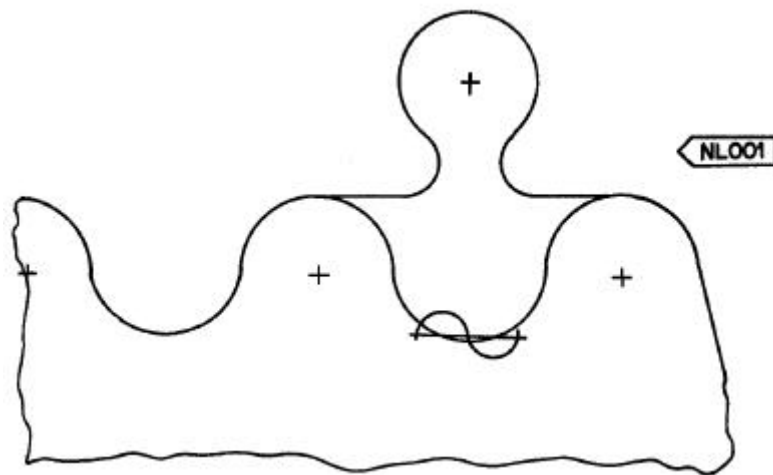


FIGURE 4: LUG INDICATED ON DRAWING

#### NOTE

If the winding shape must remain after a lug cutout, a free note (NL XXX) shall be added on drawing. See example in Figure 5.



NL 001 – The winding shape must remain after lug cutout.

FIGURE 5: EXAMPLE OF FREE NOTE INDICATION FOR WINDING SHAPE REMAIN

#### b2) Lug Not Indicated on Drawing

In this case the lug is only made to facilitate the manufacturing process and is added to the part according to production necessity. See example in Figure 6.

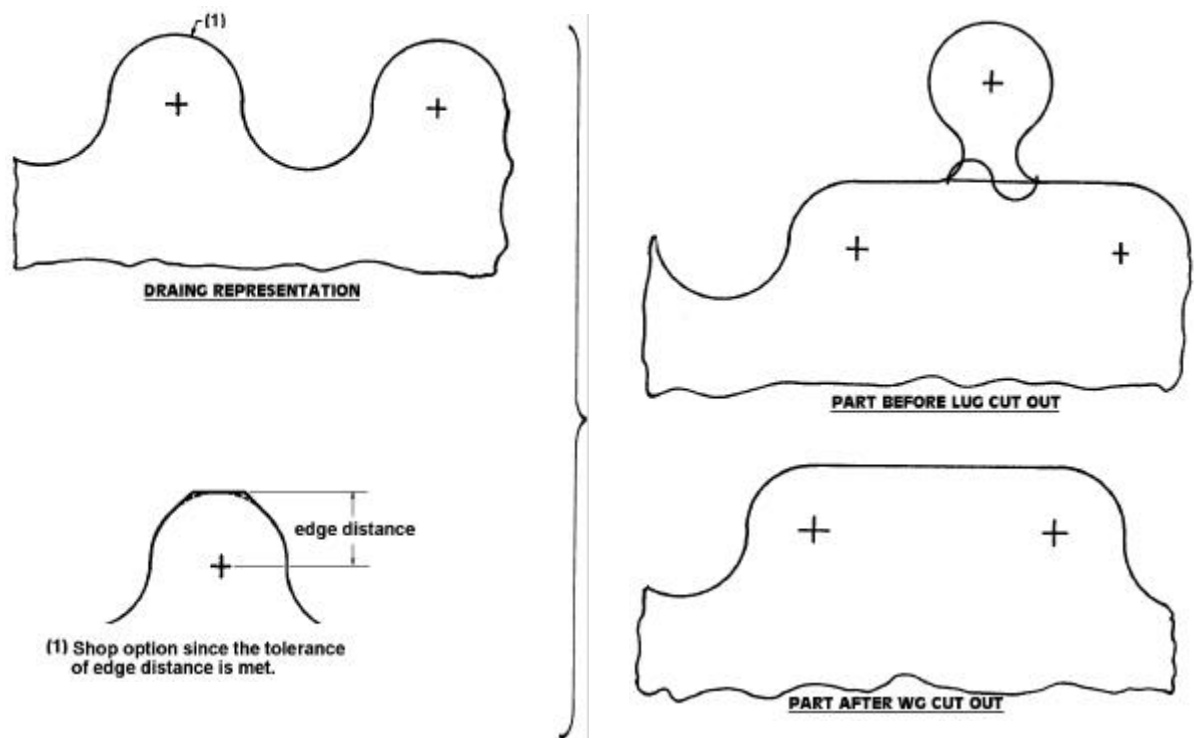


FIGURE 6: LUG NOT INDICATED ON DRAWING

### 5.1.6 GRAIN FLOW DIRECTION

Grain flow direction tolerance (general or specific) is applicable to machined parts and sheets, made of laminated, forged, or extruded material.

Consider the following cases:

- a) drawing does not indicate grain direction, but calls out NI383 (refer to item 6). In this case, grain direction is optional and may be established according to manufacturing convenience;
- b) drawing indicates grain direction but does not indicate specific angular tolerance. In this case, general angular tolerance to be considered shall be  $\pm 15^{\circ}$ ;  
Example: grain direction
- c) drawing indicates grain direction and a specific angular tolerance. In this case, indicated tolerance is mandatory.  
Example: grain direction  $\pm 10^{\circ}$

### 5.2 ASSEMBLY TOLERANCES

Assembly tolerances are applicable to the installation of components in the next higher rigid assemblies put together. They are divided into two groups:

- a) assembly tolerances which depend on tolerances of detail parts. These tolerances correspond to the sum of dimension tolerances of detail parts, which compose the final positioning of a component in an assembly. Owing to their nature, it is impossible to standardize this group of tolerances; thus, engineering is charged with the task of applying tolerances (general or specific), to the detail parts, so as to meet the requirements of the completed assembly;
- b) assembly tolerances which do not depend on the tolerances of detail parts.  
These tolerances may be specific or general. The specific tolerances apply to installations where high precision is required, such as those of the engine mount, flight control surfaces, etc, to guarantee function and/or interchangeability. General tolerances apply to medium (refer to 5.2.1) and low precision installations (refer to 5.2.2).

CAUTION

- Tolerances listed in the following items are applicable only when the case into consideration does not fit any of the mandatory cases listed in 5.3 (Table 7), mainly with regard to tolerances of symmetry (alignment) or aircraft aerodynamic smoothness.
- Reference dimensions need not to be checked when the assembly is inspected. They are represented by the dimension, followed by "REF". They have no tolerance.

5.2.1 MEDIUM PRECISION INSTALLATIONS (CLASS 1)

These tolerances apply, mainly, to the installation of components in fuselage, wings, and tail structural components, such as determining the distance between frames and stiffeners.

TABLE 6 : GENERAL TOLERANCE - MEDIUM PRECISION INSLTALLATION

| NOMINAL LINEAR<br>DIMENSION "D"<br>mm  | TOLERANCE<br>mm  | NOMINAL ANGULAR<br>DIMENSION "θ"<br>(INCLUDING 90°, NOT LISTED) | TOLERANCE<br>REFER TO 5.3 |
|---------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|-----------------------------------------------------------------|---------------------------|
| - D ≤ 1000                                                                                                                | ± 1                                                                                                 | 0° < φ < 360°                                                   | ± 0° : 30'                |
| 1000 < D ≤ 1600                                                                                                           | ± 1,6                                                                                               |                                                                 |                           |
| 1600 < D ≤ 2500                                                                                                           | ± 2                                                                                                 |                                                                 |                           |
| 2500 < D ≤ 4000                                                                                                           | ± 3                                                                                                 |                                                                 |                           |
| 4000 < D ≤ 6000                                                                                                           | ± 4                                                                                                 |                                                                 |                           |
| 6000 < D ≤ 10000                                                                                                          | ± 5                                                                                                 |                                                                 |                           |

 For each 2000 mm over 10000 mm, add ± 1 mm to the tolerance.

5.2.2 LOW PRECISION INSTALLATIONS (CLASS 2)

These tolerances apply to the installation of components in assemblies which require low precision, such as supports for wiring or lamps, light switches, access doors, passenger seats. Their values shall be two times those of the tolerances specified in Table 6. Refer to 5.3 and 6.1.

### 5.3 MANDATORY STANDARDS

The general tolerances for the items included in the subjects listed in Table 7, shall be in accordance with the relevant mandatory standards specified therein, which take precedence over this Standard and shall be called out in the production drawing.

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

NOTE: (DP) = DETAIL PART; (A) = ASSEMBLY

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                                                                                                                | MANDATORY STANDARDS                           |
|------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------|
| A    | ASSEMBLY AIRCRAFT - SEE ALSO ITEM "P"                                                                                                                                           | -                                             |
| -    | AERODYNAMIC SMOOTHNESS - REQUIREMENTS FOR EMB-110/111/120/121/312                                                                                                               | NE 02-024<br>(A)                              |
| -    | AERODYNAMIC SMOOTHNESS - REQUIREMENTS FOR CBA-123                                                                                                                               | NE 02-038<br>(A)                              |
| B    | MACHINING (AND CHEMICAL MILLING)<br>SEE ALSO ITEMS "C", "H", "J" AND "N"                                                                                                        | -                                             |
| -    | MACHINING BETWEEN CENTRES : DIMENSIONS<br>OF CENTRE HOLES                                                                                                                       | NE 03-001<br>(DP)                             |
| -    | CHEMICAL MILLING OF ALUMINUM ALLOYS: LINEAR<br>DIMENSIONS, SURFACE ROUGHNESS, OVERHANGS,<br>CONTOUR DEVIATIONS, ROUNDING OFF OF EDGES,<br>GROOVES, RECESSES, CAVITIES, POROSITY | NE 41-013<br>(DP)                             |
| C    | FASTENERS - SEE ALSO ITEM "J"                                                                                                                                                   | -                                             |
| -    | RIVETS: DRILLING, COUNTERSINKING, AND DIMPLING                                                                                                                                  | NE 03-006<br>(DP) AND (A)                     |
| -    | DRILLING, COUNTERSINKING AND DIMPLING OF HOLES<br>FOR BOLTS AND SCREWS                                                                                                          | NE 03-008<br>AND NE 03-068<br>(DP) AND (A)    |
| -    | BOLTS, SCREWS, NUTS, AND WASHERS - INSTALLATION                                                                                                                                 | NE 06-003<br>AND<br>NE 06-018<br>(DP) AND (A) |
| -    | RIVETS, "HI-LOK": INSTALLATION                                                                                                                                                  | NE 06-004<br>(DP) AND (A)                     |

continued

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                     | MANDATORY STANDARDS       |
|------|----------------------------------------------------------------------|---------------------------|
| -    | SOLID RIVETS - INSTALLATION                                          | NE 06-005<br>(A)          |
| -    | HI-TIGUE RIVETS - INSTALLATION                                       | NE 06-012                 |
| -    | BRILES RIVETS - INSTALLATION                                         | NE 06-013<br>(DP) AND (A) |
| -    | THREADED INSERT - MS51836 - INSTALLATION                             | NE 06-019<br>(A)          |
| -    | INSERT K8001 - INSTALLATION                                          | NE 06-022<br>(A)          |
| -    | LOCKBOLT RIVETS - INSTALLATION                                       | NE 06-023<br>(A)          |
| -    | HI-LITE RIVETS - INSTALLATION                                        | NE 06-024<br>(A)          |
| -    | BLIND FASTENERS - INSTALLATION                                       | NE 06-025<br>(A)          |
| -    | MFPN SERIES FLOATING NUTS - INSTALLATION                             | NE 06-027<br>(A)          |
| -    | M45932/1 THREADED INSERTS - INSTALLATION                             | NE 06-028<br>(A)          |
| -    | MS51835 - INSERTS AND STUDS - INSTALLATION                           | NE 06-029<br>(A)          |
| D    | TUBING - SEE ALSO ITEMS "I" AND "J"                                  | -                         |
| -    | TUBING - BENDING RADII                                               | NE 03-009<br>(DP)         |
| -    | NON-STRUCTURAL TUBES: DIMENSIONS AND<br>CONCENTRICITY OF FLARED ENDS | NE 04-002<br>(DP)         |
| -    | HOSE FITTING CHANNELS: HEIGHT AND POSITIONING                        | NE 04-003<br>(DP)         |
| -    | TELEFLEX CONDUIT: DIMENSIONS FOR FLARING<br>AND SWAGING              | NE 04-005<br>(DP)         |

continued

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                                                                                                                         | MANDATORY STANDARDS |
|------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| -    | AIR CONDITIONING DUCTS: DIMENSIONS FOR MANUFACTURE                                                                                                                                       | NE 04-010 (DP)      |
| -    | SLEEVE MS 21922 AND NUT MS 21921 : DIMENSIONAL LIMITATIONS WHEN FORMING TUBE FOR INSTALLATION                                                                                            | NE 06-006 (DP)      |
| -    | "PERMASWAGE" TUBES AND FITTINGS: DIMENSIONS FOR INSTALLATION                                                                                                                             | NE 06-014 (A)       |
| E    | COVERS AND DOORS - SEE ALSO ITEM "F"                                                                                                                                                     | -                   |
| -    | ACCESS DOORS: HOLE DIMENSIONS                                                                                                                                                            | NE 03-032 (DP)      |
| -    | CLEARANCES FOR ADJUSTMENT OF ACCESS PANELS, OR COVERS IN THE AIRCRAFT SKIN                                                                                                               | NE 03-061 (A)       |
| -    | EMB-121 MAIN DOOR: CLEARANCE BETWEEN ROLLER AND MECHANISM LOCK                                                                                                                           | NE 30-017 (A)       |
| -    | EMB-121 MAIN DOOR : CLEARANCE AND FREE PLAY ON ASSEMBLY                                                                                                                                  | NE 30-024 (A)       |
| F    | SHEETS AND SECTIONS - SEE ALSO ITEMS "E" AND "L"                                                                                                                                         | -                   |
| -    | ALUMINUM SHEET - OR SECTION - FORMED PARTS; THICKNESS REPRODUCTION, WAVINESS, BEND RADII, SURFACE FINISH, WRINKLES, BULGING, ANGLE DEVIATIONS, OVERHANGS, BOSSES, RECESS AND DEPRESSIONS | NE 03-040 (DP)      |
| G    | COMPOSITE MATERIALS - SEE ALSO ITEM "P"                                                                                                                                                  | -                   |
| -    | COMPOSITE STRUCTURAL LAMINATE: 100° COUNTERSINK DRILLING                                                                                                                                 | NE 03-044 (DP)      |
| -    | REINFORCED PLASTIC PARTS: THICKNESS, HOLE DIMENSIONS, RADII, ANGLES AND GENERAL DIMENSIONS                                                                                               | NE 03-047 (DP)      |
| -    | ELECTRICAL INSULATION OF COMPONENTS WITH EPOFLEX-SS COMPOUND                                                                                                                             | NE 43-002 (DP)      |

continued

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                      | MANDATORY STANDARDS |
|------|---------------------------------------------------------------------------------------|---------------------|
| H    | PANELS                                                                                | -                   |
| -    | INSTRUMENT PANELS: HOLE DIMENSIONS                                                    | NE 03-048<br>(DP)   |
| I    | CAST, FORGED AND EXTRUDED PARTS                                                       | -                   |
| -    | FORGED PARTS: DIMENSIONS AND STRAIGHTNESS                                             | NE 05-001<br>(DP)   |
| -    | RODS, WIRES, TUBES AND SPECIAL EXTRUDED<br>SECTIONS: DIMENSIONS AND FLATNESS          | NE 05-002<br>(DP)   |
| -    | CAST PARTS: DIMENSIONS, FLATNESS, STRAIGHTNESS<br>AND OUT-OF-ROUNDNESS                | NE 05-003<br>(DP)   |
| J    | THREADS - SEE ALSO ITEMS "B" AND "C"                                                  | -                   |
| -    | ACME THREADS: DIMENSIONS FOR MANUFACTURE                                              | NE 06-009<br>(DP)   |
| -    | UNIFIED THREADS: DIMENSIONS FOR MANUFACTURE                                           | NE 06-010<br>(DP)   |
| -    | ANPT THREADS: DIMENSIONS FOR MANUFACTURE                                              | NE 06-011<br>(DP)   |
| L    | LIGHTENING HOLES, CHANNELS, RECESSES, AND<br>SLOTS IN SHEET PARTS - SEE ALSO ITEM "F" | -                   |
| -    | REINFORCED BEAD DIMENSIONS                                                            | NE 15-001<br>(DP)   |
| -    | REINFORCED OVAL LIGHTENING HOLES - DIMENSIONS                                         | NE 15-304<br>(DP)   |
| -    | ROUND LIGHTENING HOLES: RECESS DIMENSIONS<br>ADDED TO SHEET THICKNESS                 | NE 15-305<br>(DP)   |
| -    | STRAIGHT REINFORCING CHANNELS: DIMENSIONS FOR<br>MANUFACTURE                          | NE 15-306<br>(DP)   |
| -    | RECESS ON ALUMINIUM ALLOY SHEET - DIMENSION                                           | NE 15-307<br>(DP)   |

continued

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                                                     | MANDATORY STANDARDS                         |
|------|----------------------------------------------------------------------------------------------------------------------|---------------------------------------------|
| -    | ROUND STRUCTURAL LIGHTENING HOLE: RIM HEIGHT                                                                         | NE 15-308<br>(DP)                           |
| -    | RECESSES IN EXTRUDED OR PRE-FORMED SECTIONS:<br>DIMENSIONS FOR MANUFACTURE                                           | NE 15-309<br>(DP)                           |
| -    | LIGHTENING CHANNELS FOR CONVEX RIMS: DIMENSIONS<br>FOR MANUFACTURE                                                   | NE 15-311<br>(DP)                           |
| -    | SLOT FOR CARRYING THROUGH REINFORCING SECTIONS:<br>DIMENSIONS FOR MANUFACTURE                                        | NE 15-312<br>(DP)                           |
| -    | REINFORCEMENT AND SLOT FOR CARRYING THROUGH<br>REINFORCING SECTIONS AT AN ANGLE: DIMENSIONS<br>FOR MANUFACTURE       | NE 15-434<br>(DP)                           |
| -    | REINFORCEMENT AND SLOT FOR CARRYING THROUGH<br>REINFORCING SECTIONS NOT AT ANGLE: DIMENSIONS<br>FOR MANUFACTURE      | NE 15-435<br>(DP)                           |
| -    | CURVED REINFORCING CHANNELS: DIMENSIONS<br>FOR MANUFACTURE                                                           | NE 15-436<br>(DP)                           |
| M    | STEEL CABLE CONTROL MECHANISMS                                                                                       | -                                           |
| -    | THREAD LENGTHS OF STEEL TABLE TURNBUCKLES,<br>FREE PLAY VALUES BETWEEN CONTROL ELEMENTS AND<br>STEEL CABLE ALIGNMENT | EMB-110/111<br>NE 30-010<br>(A)             |
|      |                                                                                                                      | EMB-121<br>NE 30-014<br>(A)                 |
| -    | THREAD LENGTHS OF STEEL CABLE TURBUCKLES,<br>LENGTH OF GRIPS IN TERMINALS                                            | EMB-710/711/<br>720/721<br>NE 30-003<br>(A) |
|      |                                                                                                                      | EMB-810<br>NE 30-002<br>(A)                 |
|      |                                                                                                                      | EMB-820<br>NE 30-009<br>(A)                 |

continued

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                              | MANDATORY STANDARDS                   |
|------|-----------------------------------------------------------------------------------------------|---------------------------------------|
| -    | TERMINAL-CABLE ASSEMBLIES: SWAGING AND TESTS                                                  | NE 30-008<br>(A)                      |
| N    | GROOVED (SHAFTS), HOUSINGS FOR BEARINGS, BALL JOINTS AND PULLEYS - SEE ALSO ITEMS "B" AND "M" | -                                     |
| -    | STEEL CABLE SECTOR GROOVE: DIMENSIONS FOR MANUFACTURE                                         | NE 30-011<br>(DP)                     |
| -    | GROOVED BEARINGS: RETENTION BY STAKING                                                        | NE 31-002<br>(DP)                     |
| -    | BEARINGS: RETENTION BY STAKING                                                                | NE 31-003<br>(DP)                     |
| -    | INVOLVING GROOVED ELEMENTS: DIMENSIONS FOR MANUFACTURE                                        | NE 32-001<br>AND<br>NE 32-006<br>(DP) |
| -    | GROOVED SAE J 500 3/8 in: DIMENSIONS FOR MANUFACTURE                                          | NE 32-002<br>(DP)                     |
| P    | PROTECTION, FINISH, SEALING AND BONDING - SEE ALSO ITEM "A"                                   | -                                     |
| -    | STRUCTURAL BONDING: TEST SAMPLE DIMENSIONS                                                    | NE 07-008<br>(DP)                     |
| -    | RETAINING RINGS: DEFECTS IN PROTECTIVE COAT                                                   | NE 28-002<br>(DP)                     |
| -    | WASH-PRIMER: APPLICATION                                                                      | NE 40-007<br>(DP)                     |
| -    | ZINC-CHROMATE PRIMER: APPLICATION                                                             | NE 40-008<br>(A)                      |
| -    | EPOXY PAINT - APPLICATION                                                                     | NE 40-009<br>(DP)                     |
| -    | CADMIUM-PLATING BY ELECTRODEPOSITION: APPLICATION                                             | NE 40-010<br>(DP)                     |
| -    | CHROMIUM-PLATING BY ELECTRODEPOSITION: APPLICATION                                            | NE 40-015<br>(DP)                     |

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TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                       | MANDATORY STANDARDS |
|------|----------------------------------------------------------------------------------------|---------------------|
| -    | ANODIZING OF ALUMINIUM ALLOYS WITH<br>SULPHURIC ACID: APPLICATION                      | NE 40-016<br>(DP)   |
| -    | SILVER PLATING: APPLICATION                                                            | NE 40-020<br>(DP)   |
| -    | PRESSURIZED FUSELAGES-SEALING:<br>MINIMUM SEAL DIMENSIONS                              | NE 40-022<br>(A)    |
| -    | RADOMES - APPLICATION OF "ASTROCOAT"<br>SYSTEM: COAT THICKNESS                         | NE 40-025<br>(A)    |
| -    | PROCEDURES FOR INTEGRAL FUEL TANKS<br>PROTECTION AND SEALING                           | NE 40-027<br>(A)    |
| -    | NICKEL-PLATING BY ELETRODEPOSITION:<br>APPLICATION                                     | NE 40-028<br>(DP)   |
| -    | ENAMEL, POLYURETHANE FOR FINISHING COATS:<br>APPLICATION                               | NE 40-029<br>(A)    |
| -    | ANODIZING OF ALUMINUM ALLOYS WITH CHROMIC<br>ACID: APPLICATION                         | NE 40-030<br>(DP)   |
| -    | PRE-TREATMENT OF MAGNESIUM ALLOYS:<br>DIMENSIONAL CHANGES                              | NE 40-031<br>(DP)   |
| -    | STEEL, ALUMINUM-ALLOY AND MAGNESIUM-ALLOY<br>PARTS (EMB-326) - PROTECTION: APPLICATION | NE 40-033<br>(DP)   |
| -    | NONSLIP COAT - APPLICATION ON SMALL AIRCRAFT                                           | NE 40-061<br>(A)    |
| -    | SHOT PEENING: SURFACE ROUGHNESS                                                        | NE 40-072<br>(DP)   |
| -    | SMALL AIRCRAFT CABIN AND BAGGAGE COMPARTMENT<br>DOORS SEALING: APPLICATION             | NE 40-078<br>(A)    |
| Q    | WELDING                                                                                | -                   |
| -    | ALUMINUM ALLOYS: FUSION WELDING - WELDING BEAD<br>DIMENSIONS                           | NE 40-055<br>(A)    |

continued

TABLE 7 : GENERAL TOLERANCES - MANDATORY STANDARDS

continued

| ITEM | SUBJECT/ELEMENTS WITH TOLERANCES                                                                                                                                           | MANDATORY STANDARDS       |
|------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------|
| -    | CARBON STEELS AND LOW ALLOY, AND HEAT-<br>AND CORROSION-RESISTANT ALLOYS: FUSION WELDING-<br>WELDING BEAD DIMENSIONS                                                       | NE 40-056<br>(DP) AND (A) |
| -    | STAINLESS STEELS, CARBON STEELS AND ALUMINUM<br>ALLOYS WITH THICKNESS GREATER THAN 0,2 mm :<br>IMPRESSION DEPTH, WELD PENETRATION AND SHEET<br>SPACING DURING SPOT-WELDING | NE 40-073<br>(A)          |

## 6. GENERAL NOTES

### 6.1 DRAWING CALL OUT

This Standard shall be called out in the production drawings (see Note of item 2) through the following notes.

- a) NI 380 - Tolerances for primary parts and assembly class < \* >, according to NE 03-063 and indicated.

\* indicate:

< - > drawings of detail parts only,

< 1 > drawings of assemblies with medium precision installation (refer to 5.2.1),

< 2 > drawings of assemblies with low-precision installation (refer to 5.2.2);

- b) NI 383 - Grain direction not indicated is optional.

NOTE

Whenever a Standard listed in Table 7 is applicable, it should be called out in the production drawings, through its specific note.

## 7. BIBLIOGRAPHIC REFERENCES

BS 308 - "Engineering Practice - Part 2 - Dimensioning and Tolerancing of size".

SM 01-033 - "Dimensions and Form Tolerances".

SM 01-037 - "Machined Parts - Drawing Preparation, Machining Practices and Inspection".

ISO 2768 - "Permissible Machining Variations in Dimensions without Tolerance Indication".

DIN 7168 - "General Tolerances - Linear and Angular Dimensions".

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